

Dependent Censoring and its Effects on Cox Model Tests - A Simulation Study

Dylan P Hebert

Introduction

Cox proportional hazard models are a common non-parametric estimation strategy for survival data. The model quantifies the effect of unit increases in covariates on time-to-event outcomes, with an assumption that these effects are proportional and multiplicative over time. In this study, we will explore the effects of various censoring qualities on the size and power of several hypothesis tests from the cox model.

Methods

Data was simulated in an R environment using the `sim.survdata()` function from the `coxed` library, which simulates time-to-event data. The covariate values were generated from the `rbinom()` and `runif()` functions from the base R library, which generate pseudo random values from the binomial and uniform distributions, respectively. The time-to-event data was always conditioned upon three covariates fed to the `sim.survdata()` function, however we used two different simple Cox models to evaluate each setting:

$$1) h(t | Z) = h_0(t) \exp(\beta_1 Z_1)$$

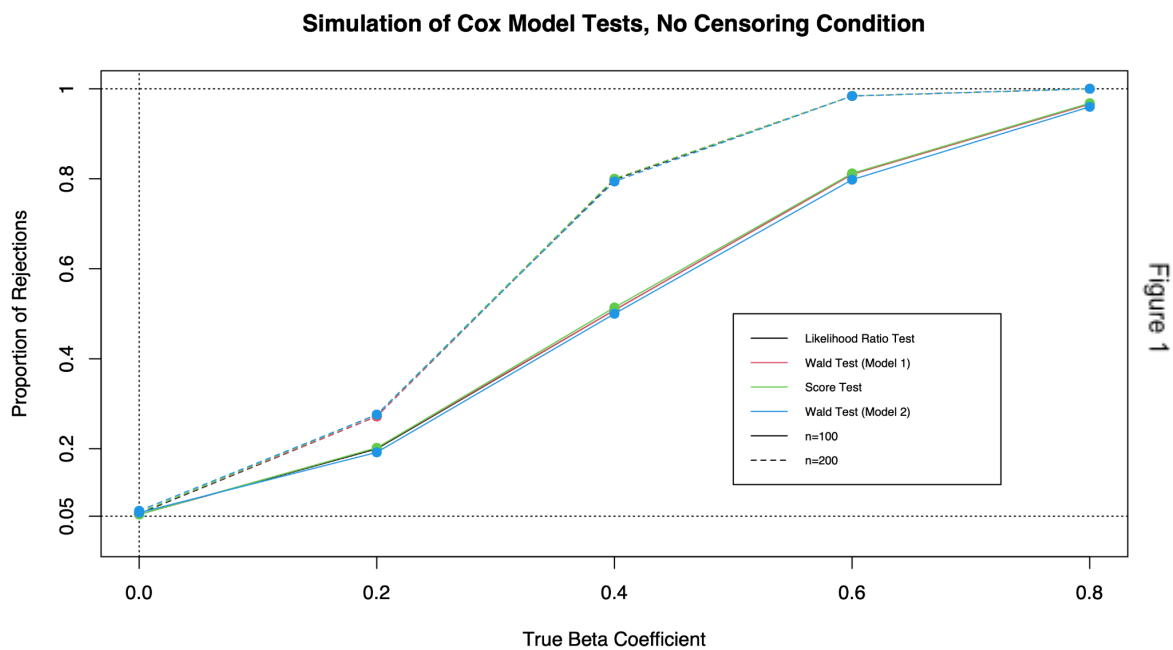
$$2) h(t | Z) = h_0(t) \exp(\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3)$$

Data was simulated under 50 different settings, with varying values for observation numbers ($n = \{100, 200\}$), the coefficient value for the first covariate ($\beta_1 = \{0, 0.2, 0.4, 0.6, 0.8\}$), the degree of censoring present among the observations, and whether or not censoring was dependent upon the covariates (No censoring, 20% independent censoring, 60% independent censoring, 20% dependent censoring, 60% independent censoring). Each permutation of these options gives a setting for the simulation. The

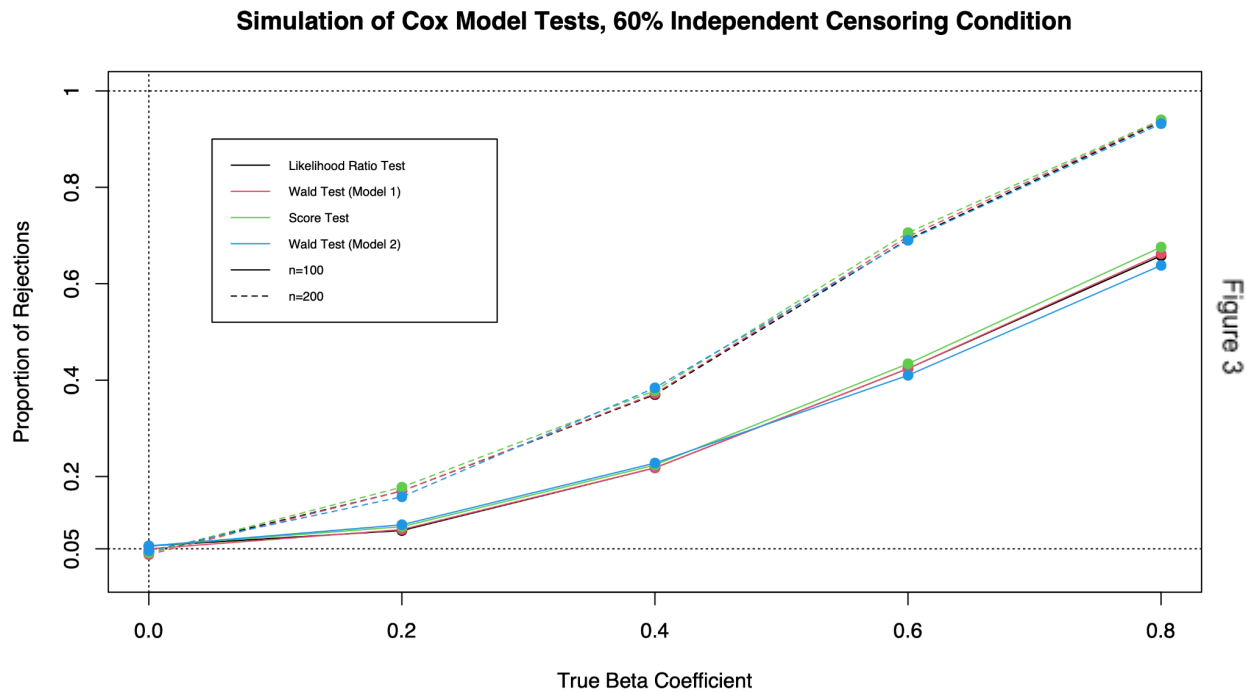
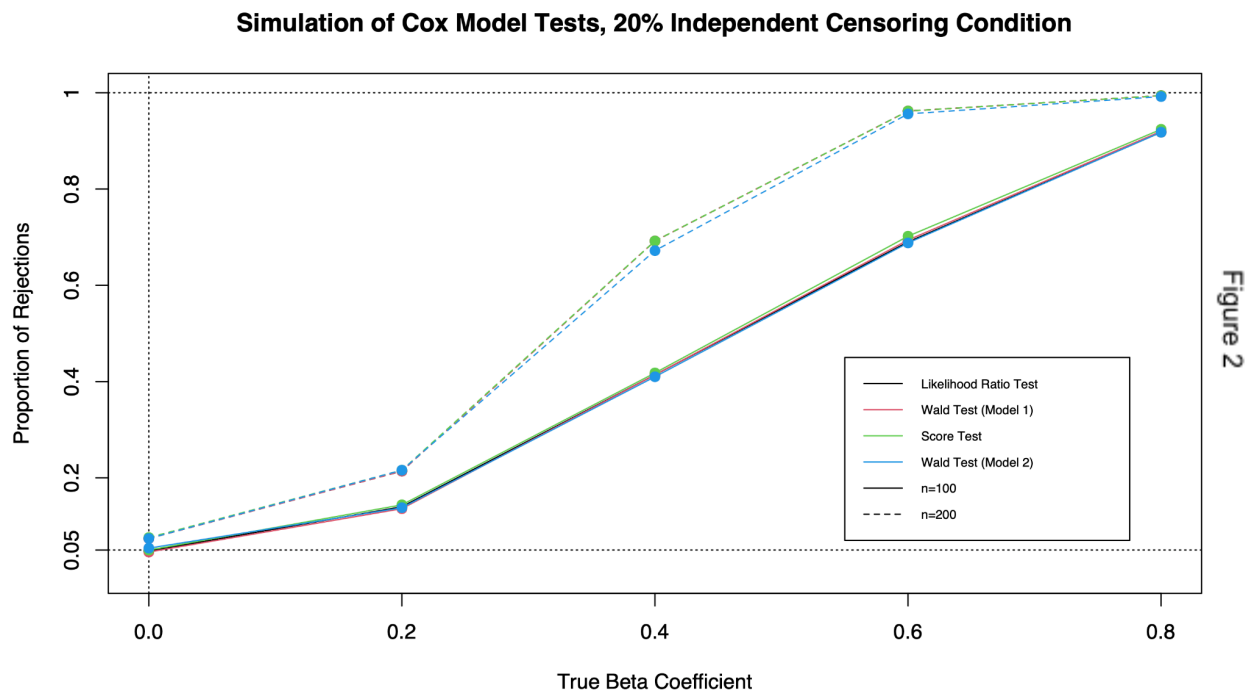
coefficient values for the second and third covariates are held constant at $\beta_2 = 0.04$ and $\beta_3 = 0.02$. Each setting was simulated 500 times and the conclusions of four hypothesis tests were saved. The tests used were the Wald, likelihood, and score tests from the first model with one covariate, and just the Wald test of the β_1 coefficient of the second model. A proportion of the 500 conclusions was calculated for each test from each setting ($\frac{Rejections}{500}$). A significance level of 0.05 was used for all test conclusions ($\alpha = 0.05$). Refer to the appendix at the end of the report for more specific details on the techniques used to simulate and manage the data.

Results

In our most basic setting of the simulation, without censoring, we see the rejection proportions start around 0.05 at an effect size of 0, and climb to nearly 1 for both sample size conditions at our highest effect size of 0.8. Tests were fairly homogenous in their results with respect to sample size.

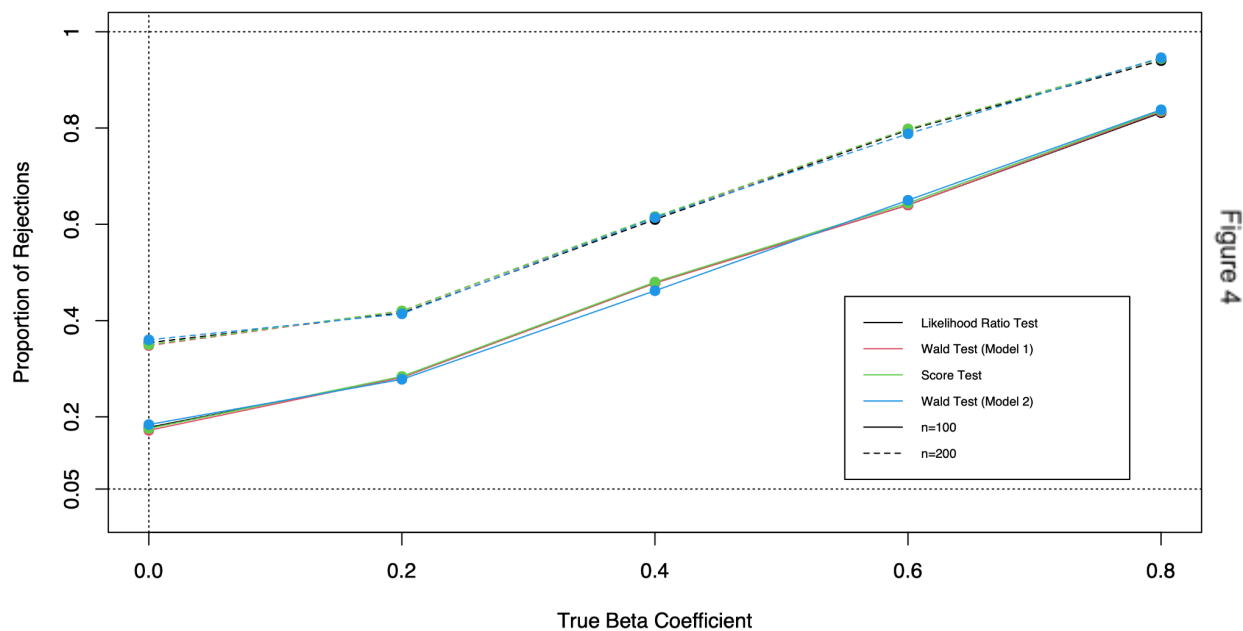


Under our independently censored conditions, we have similar findings at low effect sizes. The rejection proportion at an effect size of 0 was slightly lower than the significance level under the more heavily censored data. Rejection proportions rose significantly as effect size approached 0.08, but the $n=200$ conditioned tests did not quite reach a proportion of 1 under the more heavily censored condition. The $n=100$ conditioned tests did not make it as high under either degree of censoring. Test conclusions were still fairly homogenous here but to a lesser extent with a lower sample size and a higher degree of censoring, when compared to the simulations with no censoring.

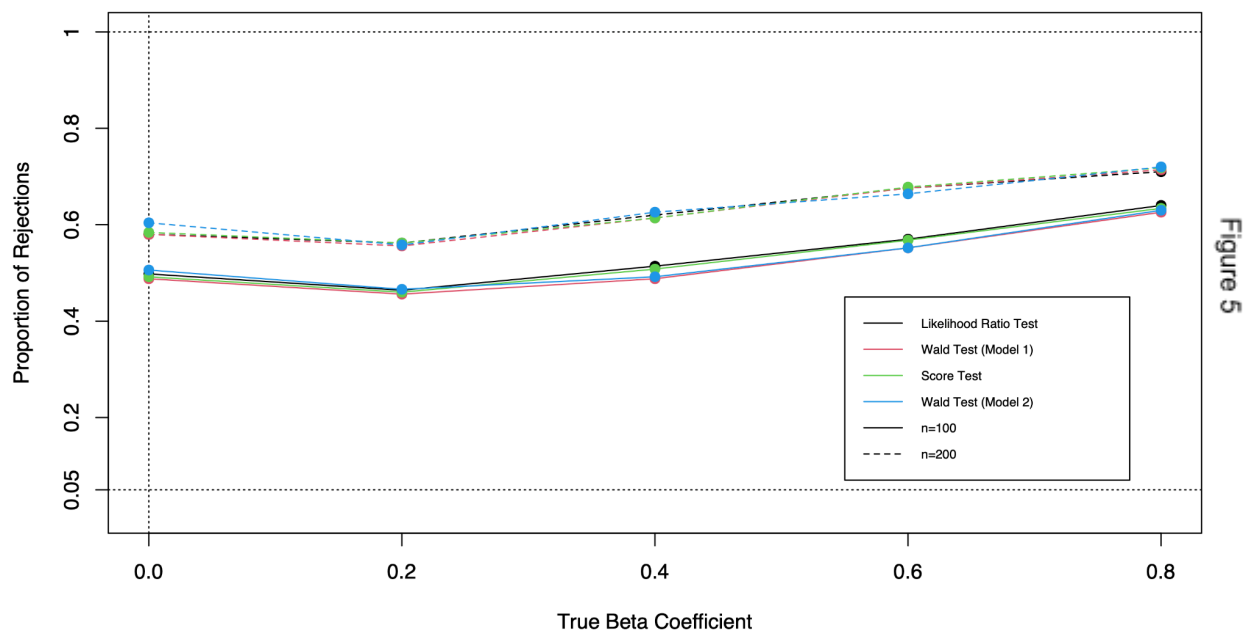


Under the dependently censored conditions, we see rejection proportions start much higher at an effect size of 0, around 0.2 and 0.38 under the lightly censored data, 0.5 and 0.6 for the more heavily censored data. The proportion of rejections was less sensitive to the effect sizes increasing, especially among the data with higher censoring. The lightly censored data did not reach 1 in either sample size condition, and the heavily censored data barely increased above its initial proportion at all. A higher sample size saw a higher proportion of rejections at all effect sizes. Tests were again mostly homogenous, less so in the simulations with higher degrees of censoring.

Simulation of Cox Model Tests, 20% Dependent Censoring Condition



Simulation of Cox Model Tests, 60% Dependent Censoring Condition



Discussion

Dependent censoring seems to present a problem for the Cox model under the conditions we tested. An ideal hypothesis test should present a few qualities, the rejection proportion being very close to our chosen significance level on average when the true beta coefficient value is 0, and increasing as quickly as possible as the true coefficient value gets higher. There is a balance here with respect to size and power, as a test must be sensitive enough to reject H_0 at small effect sizes, but still correctly accept H_0 when true effect sizes are at or near 0.

Under the settings without censoring or where censoring was independent of covariates, all of the tests exhibit this behavior very well. We see them increase in statistical power at higher sample sizes, while maintaining a good acceptance rate at an effect size of 0. Under the settings with censoring dependent upon the covariates, the tests exhibit less ideal behavior. Rejection rates were always higher than 0, even at a true effect size of 0. And while the higher sample size condition had a higher rejection rate, it was higher on the entire curve, meaning this quality could not be obviated by just using a larger sample.

Conclusion

The biggest takeaway from these results is the need for corrections when dependent censoring is suspected in a data set. Larger samples likely cannot counter the problems associated with these model tests, so a different inferential approach may be necessary, and researchers should try to be aware of ways that censoring can be influenced by factors within the model or the event in question.

For standard survival analysis where the assumption of independent censoring holds (along with confirmation of the proportional hazards assumption), the Cox proportional hazards model performs well. It should be noted that for heavy independent censoring, sample sizes requirements could be in excess of 200 to be very sure of any test conclusions. It may be wise to “pad” any sample size expectations researchers have, in case censoring ends up being heavier than expected, or to wait until a body of research develops with a larger combined sample in order to perform a meta-analysis if the effect size of the covariate in question is more modest.

Appendix

```
library(survival)
library(coxed)

# make vectors which contain values for censoring, censoring
# conditions,
# beta values, and sample sizes. similar vector indexes will contain
# values for a
# specific sim setting
cens.g <- rep(c(0,0.2,0.6,0.2,0.6), each=10)
cens.bool.g <- rep(c(F,F,F,T,T), each=10)
beta1.g <- rep(c(0,0.2,0.4,0.6,0.8), 10)
n.g <- rep(c(100,200), each=5, 5)
alpha <- .05

# initializing variables local to the loop
cens <- 0
cens.bool <- F
beta1 <- 0
n <- 0

# assign vector for inner/outer simulation loops, create vectors to
# store
# rejections. 0 indicates acceptance of H_0, 1 indicates rejection
J <- 50
K <- 500
reject.log <- numeric(length=K)
reject.wald <- numeric(length=K)
reject.score <- numeric(length=K)
reject.m.wald <- numeric(length=K)
p.log <- numeric(length=J)
p.wald <- numeric(length=J)
p.score <- numeric(length=J)
p.m.wald <- numeric(length=J)
```

```
# outer loop which assigns local parameters from our established
vector objects
for (x in 1:J) {

  cens <- cens.g[x]
  cens.bool <- cens.bool.g[x]
  beta1 <- beta1.g[x]
  n <- n.g[x]

  for (i in 1:K) {
    #simulates data
    xf <- data.frame(x1=rbinom(n,1,0.5), x2=runif(n,-0.5,0.5),
x3=rbinom(n,1,0.5))
    tmp <- sim.survdata(n, T=1000, X=xf, censor=cens,
beta=c(beta1,0.04,0.02),
                      censor.cond=cens.bool)

    #create survival, cox objects
    tmp.surv <- Surv(tmp$data$y,tmp$data$failed)
    tmp.cox <- coxph(tmp.surv~tmp$data$x1)
    tmp.coxm <- coxph(tmp.surv~tmp$data$x1+tmp$data$x2+tmp$data$x3)
    tmp.cox.summ <- summary(tmp.cox)
    tmp.coxm.summ <- summary(tmp.coxm)

    #rejection vectors
    reject.log[i] <- tmp.cox.summ$logtest[3] < alpha
    reject.wald[i] <- tmp.cox.summ$waldtest[3] < alpha
    reject.score[i] <- tmp.cox.summ$scetest[3] < alpha
    reject.m.wald[i] <- tmp.coxm.summ$coef[13] < alpha
  }

  # these vectors contain proportions of rejections from each sim
setting
  p.log[x] <- mean(as.numeric(reject.log))
  p.wald[x] <- mean(as.numeric(reject.wald))
  p.score[x] <- mean(as.numeric(reject.score))
  p.m.wald[x] <- mean(as.numeric(reject.m.wald))
}
```

```
#####
# p vectors will be indexed based on the indexes from our parameter
# vectors
# from above. a quick guide:
# -No censoring
# --Effect Sizes: (0, 0.2, 0.4, 0.6, 0.8) at n=100 and
# --          (0, 0.2, 0.4, 0.6, 0.8) at n=200
# --- indexed by 1:10, respectively
#
# -20% independent censoring
# --Effect Sizes: (0, 0.2, 0.4, 0.6, 0.8) at n=100 and
# --          (0, 0.2, 0.4, 0.6, 0.8) at n=200
# --- indexed by 11:20, respectively
#
# -60% independent censoring
# --Effect Sizes: (0, 0.2, 0.4, 0.6, 0.8) at n=100 and
# --          (0, 0.2, 0.4, 0.6, 0.8) at n=200
# --- indexed by 21:30, respectively
#
# -20% dependent censoring
# --Effect Sizes: (0, 0.2, 0.4, 0.6, 0.8) at n=100 and
# --          (0, 0.2, 0.4, 0.6, 0.8) at n=200
# --- indexed by 31:40, respectively
#
# -60% dependent censoring
# --Effect Sizes: (0, 0.2, 0.4, 0.6, 0.8) at n=100 and
# --          (0, 0.2, 0.4, 0.6, 0.8) at n=200
# --- indexed by 41:50, respectively
#####
# from this, we can extract the values into matrices to make plots
# with
rownames <- c("Beta=0", "Beta=0.2", "Beta=0.4", "Beta=0.6",
"Beta=0.8")
effect.size <- c(0,0.2,0.4,0.6,0.8)
no.censoring <- cbind(p.log[1:5], p.log[6:10], p.wald[1:5],
p.wald[6:10],
                    p.score[1:5], p.score[6:10], p.m.wald[1:5],
p.m.wald[6:10])
ind.20.censoring <- cbind(p.log[11:15], p.log[16:20], p.wald[11:15],
```

```

p.wald[16:20],
                                p.score[11:15], p.score[16:20],
p.m.wald[11:15], p.m.wald[16:20])
ind.60.censoring <- cbind(p.log[21:25], p.log[26:30], p.wald[21:25],
p.wald[26:30],
                                p.score[21:25], p.score[26:30],
p.m.wald[21:25], p.m.wald[26:30])
dep.20.censoring <- cbind(p.log[31:35], p.log[36:40], p.wald[31:35],
p.wald[36:40],
                                p.score[31:35], p.score[36:40],
p.m.wald[31:35], p.m.wald[36:40])
dep.60.censoring <- cbind(p.log[41:45], p.log[46:50], p.wald[41:45],
p.wald[46:50],
                                p.score[41:45], p.score[46:50],
p.m.wald[41:45], p.m.wald[46:50])

# and build plots from these matrices
legend <- c("Likelihood Ratio Test", "Wald Test (Model 1)", "Score
Test", "Wald Test (Model 2)", "n=100", "n=200")

# no censoring condition, figure 1
plot(effect.size, no.censoring[,1],
      main="Simulation of Cox Model Tests, No Censoring Condition",
      ylab="Proportion of Rejections", xlab="True Beta
Coefficient", ylim=0:1,
      type="o", pch = 19, yaxt="n", col=1, lty=1)
axis(side=2, at=c(0.05, 0.2, 0.4, 0.6, 0.8, 1), labels = c("0.05",
"0.2", "0.4", "0.6", "0.8", "1"))
abline(h=c(0.05,1),v=0,lty=3)
lines(effect.size, no.censoring[,2], type="o", pch = 19, col=1,
lty=2)
lines(effect.size, no.censoring[,3], type="o", pch = 19, col=2,
lty=1)
lines(effect.size, no.censoring[,4], type="o", pch = 19, col=2,
lty=2)
lines(effect.size, no.censoring[,5], type="o", pch = 19, col=3,
lty=1)
lines(effect.size, no.censoring[,6], type="o", pch = 19, col=3,
lty=2)

```

```

lines(effect.size, no.censoring[,7], type="o", pch = 19, col=4,
lty=1)
lines(effect.size, no.censoring[,8], type="o", pch = 19, col=4,
lty=2)
legend(0.5,0.5,cex=.6,legend=legend,lty=c(1,1,1,1,1,2),col=c(1,2,3,4,
1,1))

# 20% independent censoring condition, figure 2
plot(effect.size, ind.20.censoring[,1],
      main="Simulation of Cox Model Tests, 20% Independent Censoring
Condition",
      ylab="Proportion of Rejections", xlab="True Beta Coefficient",
ylim=0:1,
      type="o", pch = 19, yaxt="n", col=1, lty=1)
axis(side=2, at=c(0.05, 0.2, 0.4, 0.6, 0.8, 1), labels = c("0.05",
"0.2", "0.4", "0.6", "0.8", "1"))
abline(h=c(0.05,1),v=0,lty=3)
lines(effect.size, ind.20.censoring[,2], type="o", pch = 19, col=1,
lty=2)
lines(effect.size, ind.20.censoring[,3], type="o", pch = 19, col=2,
lty=1)
lines(effect.size, ind.20.censoring[,4], type="o", pch = 19, col=2,
lty=2)
lines(effect.size, ind.20.censoring[,5], type="o", pch = 19, col=3,
lty=1)
lines(effect.size, ind.20.censoring[,6], type="o", pch = 19, col=3,
lty=2)
lines(effect.size, ind.20.censoring[,7], type="o", pch = 19, col=4,
lty=1)
lines(effect.size, ind.20.censoring[,8], type="o", pch = 19, col=4,
lty=2)
legend(0.55,0.45,cex=.6,legend=legend,lty=c(1,1,1,1,1,2),col=c(1,2,3,
4,1,1))

# 60% independent censoring condition, figure 3
plot(effect.size, ind.60.censoring[,1],
      main="Simulation of Cox Model Tests, 60% Independent Censoring
Condition",
      ylab="Proportion of Rejections", xlab="True Beta Coefficient",

```

```

ylim=0:1,
  type="o", pch = 19, yaxt="n", col=1, lty=1)
axis(side=2, at=c(0.05, 0.2, 0.4, 0.6, 0.8, 1), labels = c("0.05",
"0.2", "0.4", "0.6", "0.8", "1"))
abline(h=c(0.05,1),v=0,lty=3)
lines(effect.size, ind.60.censoring[,2], type="o", pch = 19, col=1,
lty=2)
lines(effect.size, ind.60.censoring[,3], type="o", pch = 19, col=2,
lty=1)
lines(effect.size, ind.60.censoring[,4], type="o", pch = 19, col=2,
lty=2)
lines(effect.size, ind.60.censoring[,5], type="o", pch = 19, col=3,
lty=1)
lines(effect.size, ind.60.censoring[,6], type="o", pch = 19, col=3,
lty=2)
lines(effect.size, ind.60.censoring[,7], type="o", pch = 19, col=4,
lty=1)
lines(effect.size, ind.60.censoring[,8], type="o", pch = 19, col=4,
lty=2)
legend(0.05,0.9,cex=.6,legend=legend,lty=c(1,1,1,1,1,2),col=c(1,2,3,4
,1,1))

# 20% dependent censoring condition, figure 4
plot(effect.size, dep.20.censoring[,1],
  main="Simulation of Cox Model Tests, 20% Dependent Censoring
Condition",
  ylab="Proportion of Rejections", xlab="True Beta Coefficient",
ylim=0:1,
  type="o", pch = 19, yaxt="n", col=1, lty=1)
axis(side=2, at=c(0.05, 0.2, 0.4, 0.6, 0.8, 1), labels = c("0.05",
"0.2", "0.4", "0.6", "0.8", "1"))
abline(h=c(0.05,1),v=0,lty=3)
lines(effect.size, dep.20.censoring[,2], type="o", pch = 19, col=1,
lty=2)
lines(effect.size, dep.20.censoring[,3], type="o", pch = 19, col=2,
lty=1)
lines(effect.size, dep.20.censoring[,4], type="o", pch = 19, col=2,
lty=2)
lines(effect.size, dep.20.censoring[,5], type="o", pch = 19, col=3,

```

```

lty=1)
lines(effect.size, dep.20.censoring[,6], type="o", pch = 19, col=3,
lty=2)
lines(effect.size, dep.20.censoring[,7], type="o", pch = 19, col=4,
lty=1)
lines(effect.size, dep.20.censoring[,8], type="o", pch = 19, col=4,
lty=2)
legend(0.55,0.45,cex=.6,legend=legend,lty=c(1,1,1,1,1,2),col=c(1,2,3,
4,1,1))

# 60% dependent censoring condition, figure 5
plot(effect.size, dep.60.censoring[,1],
      main="Simulation of Cox Model Tests, 60% Dependent Censoring
Condition",
      ylab="Proportion of Rejections", xlab="True Beta Coefficient",
ylim=0:1,
      type="o", pch = 19, yaxt="n", col=1, lty=1)
axis(side=2, at=c(0.05, 0.2, 0.4, 0.6, 0.8, 1), labels = c("0.05",
"0.2", "0.4", "0.6", "0.8", "1"))
abline(h=c(0.05,1),v=0,lty=3)
lines(effect.size, dep.60.censoring[,2], type="o", pch = 19, col=1,
lty=2)
lines(effect.size, dep.60.censoring[,3], type="o", pch = 19, col=2,
lty=1)
lines(effect.size, dep.60.censoring[,4], type="o", pch = 19, col=2,
lty=2)
lines(effect.size, dep.60.censoring[,5], type="o", pch = 19, col=3,
lty=1)
lines(effect.size, dep.60.censoring[,6], type="o", pch = 19, col=3,
lty=2)
lines(effect.size, dep.60.censoring[,7], type="o", pch = 19, col=4,
lty=1)
lines(effect.size, dep.60.censoring[,8], type="o", pch = 19, col=4,
lty=2)
legend(0.55,0.45,cex=.6,legend=legend,lty=c(1,1,1,1,1,2),col=c(1,2,3,
4,1,1))

```